2. Make the HFSS dataset supervise an inverse request

HFSS supplies (geometry, substrate) → absorption spectrum, while the desired ML task is (band, threshold) → geometry. The missing step is deterministic target-label construction from the simulated catalogue.

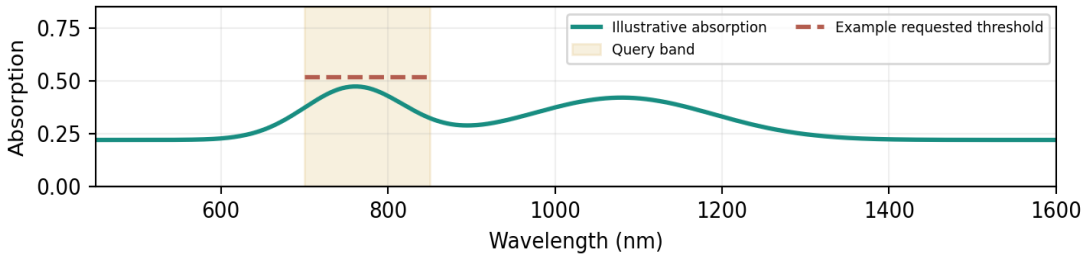


Figure 1. Example showing how a simulated spectrum is assessed against a requested threshold in a wavelength band. Entirely illustrative; not a repository prediction.

$$B(q) = \{\lambda_i : a \le \lambda_i \le b\}$$

$$J_j(q) = \frac{1}{|B|} \sum_{\lambda_i \in B} \max(0, X - A_j(\lambda_i))^2$$

$$j^*(q) = \arg\min_{j \in C(q)} [J_j(q) + \varepsilon P_j]$$

For each synthetic request q, evaluate the stored, validated HFSS catalogue on its 1 nm reporting grid, with interpolated band endpoints where needed. C(q) is the fixed catalogue with any operating constraints. J is zero when every resolved in-band wavelength meets the threshold. P_j is a fixed tie-breaker, such as normalized gold area fraction. The catalogue argmin is an offline label reduction, not an inference-time optimization or an additional learned model. Narrow-band labels inherit the actual HFSS spectral resolution limits.

Training queries

- Generate 10,000 fixed-seed synthetic requests q with continuous 450 ≤ a < b ≤ 1600 nm and 0 ≤ X ≤ 1. Include very narrow and wide bands rather than imposing a 20 nm minimum; flag bands narrower than validated spectral detail as uncertain.
- Include both achievable and unmet requests. Assign the best verified catalogue entry even if its objective J is nonzero; mark feasibility separately.
- Use a balanced request distribution across short, middle and long wavelength intervals, and across demanding/less-demanding thresholds.
- Split geometries before query-label construction to prevent spectra from the same geometry leaking into held-out evaluation.

The 10,000 requests are derived examples from 160 full-wave spectra, not independent full-wave data. Only geometry-partitioned spectra provide genuinely held-out physical designs. For very narrow bands, error depends on the true solved-frequency density, not merely the 1 nm export grid.